

Sophie Huiberts

Curriculum Vitae, May 2026

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Work experience

- Oct 2023 – present **Chargée de Recherche**, *CNRS, LIMOS laboratory, Clermont-Ferrand*
- Jul 2022 – Aug 2023 **Postdoctoral fellow**, *Computer Science Department, Columbia University*
- Dec 2017 – Jun 2022 **PhD candidate**, *Centrum Wiskunde & Informatica*
- Supervisor: Daniel Dadush. Promotor: Gunther Cornelissen
 - Defense date: May 16th 2022
 - Thesis: Geometric Aspects of Linear Programming: Shadow Paths, Central Paths, and a Cutting Plane Method (Stieltjes Prize, Gijs de Leve Prize)
- Feb 2017 – Nov 2017 **Research trainee**, *Centrum Wiskunde & Informatica*
- Supervisor: Daniel Dadush. Subject: smoothed analysis of the simplex method
- Sep 2016 – Aug 2017 **Student board member**, *Department of Mathematics, Utrecht University*
- Representing students' interests on education and policy matters

Education

- Feb 2016 – Jan 2018 **Master Mathematical Sciences**, *Utrecht University*
- Sep 2012 – Aug 2017 **Bachelor Computer Science**, *Utrecht University*
- Sep 2012 – Feb 2016 **Bachelor Mathematics**, *Utrecht University*

Honours and Awards

- Jan 2026 **Meritorious Service Award**, *Mathematical Programming*, 2025
- Jan 2024 **Gijs de Leve Prize**, *Landelijk Netwerk Mathematische Besliskunde*, best Dutch PhD thesis in mathematics of operations research, 2020–2023
- Apr 2023 **Stieltjes Prize**, *Platform Wiskunde Nederland*, best Dutch PhD thesis in mathematics, academic year 2021–2022
- Jun 2021 **Rising Star Speaker**, *TCS Women Spotlight Workshop*, ACM Symposium on Theory of Computing (STOC)
- Sep 2019 **Attendee**, *Heidelberg Laureate Forum*, selective conference admitting 100 young researchers across mathematics and computer science

Grants and Fellowships

- 2024 – 2028 **ANR JCJC Grant**, *Agence Nationale de la Recherche*, Project: Towards Testable Theories of Linear Programming
- 2022 – 2023 **Junior Fellowship**, *Simons Society of Fellows*

Featured in

- *Researchers Discover the Optimal Way To Optimize*, Steve Nadis, Quanta Magazine, October 2026. Syndicated in Wired, December 2026.
- *Women in Optimization: Dr. Sophie Huiberts*, Elisabeth Rodriguez Heck, Gurobi Optimization, July 2024.

- *Proof by example*: Sophie Huiberts, Clara Stegehuis and Francesca Arici, Nieuw Archief voor Wiskunde, December 2022.
- *Mathématiques: lumière sur les étranges performances de l'algorithme du simplexe*, Clémentine Laurens, Le Monde, 4 October 2022.
- *Het diversiteitsprobleem is geen vrouwenprobleem*, Ans Hekkenberg, CWI 75th anniversary special of New Scientist, 2021.

Publications

In my field of research, authors are listed in alphabetical order.

- [1] E. Bach, Y. Disser, S. Huiberts, and N. Mosis, "An unconditional lower bound for the active-set method in convex quadratic maximization," in *Proceedings of the 2026 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, Society for Industrial and Applied Mathematics, Jan. 2026, pp. 314–327. DOI: 10.1137/1.9781611978971.14 arXiv: 2507.16648.
- [2] G. Bonnet, D. Dadush, U. Grupel, S. Huiberts, and G. Livshyts, "Asymptotic bounds on the combinatorial diameter of random polytopes," *Discrete and Computational Geometry*, Feb. 2026, preliminary version in SoCG 2022, ISSN: 1432-0444. DOI: 10.1007/s00454-025-00814-6
- [3] E. Bach, A. E. Black, S. Huiberts, and S. Kafer, *Beyond smoothed analysis: Analyzing the simplex method by the book*, to appear in STOC 2026, 2025. arXiv: 2510.21613.
- [4] E. Bach and S. Huiberts, "Optimal smoothed analysis of the simplex method," in *66th IEEE Symposium on Foundations of Computer Science (FOCS)*, 2025, pp. 1832–1859. arXiv: 2504.04197.
- [5] S. Huiberts, Y. T. Lee, and X. Zhang, "Upper and lower bounds on the smoothed complexity of the simplex method," *TheoretCS*, vol. 4, Oct. 2025, Art. no. 23, preliminary version in STOC 2023. DOI: 10.46298/theoretics.25.23 arXiv: 2211.11860.
- [6] S. Borst, D. Dadush, S. Huiberts, and D. Kashaev, "A nearly optimal randomized algorithm for explorable heap selection," *Mathematical Programming*, vol. 210, no. 1–2, pp. 75–96, Nov. 2024, preliminary version in IPCO 2023. DOI: 10.1007/s10107-024-02145-5 arXiv: 2210.05982.
- [7] D. Dadush, C. Hojny, S. Huiberts, and S. Weltge, "A simple method for convex optimization in the oracle model," *Mathematical Programming*, Aug. 2023, preliminary version in IPCO 2022. DOI: 10.1007/s10107-023-02005-8 arXiv: 2011.08557.
- [8] D. Dadush, S. Huiberts, B. Natura, and L. A. Végh, "A scaling-invariant algorithm for linear programming whose running time depends only on the constraint matrix," *Mathematical Programming*, Apr. 2023, preliminary version in STOC 2020. DOI: 10.1007/s10107-023-01956-2 arXiv: 1912.06252.
- [9] S. Borst, D. Dadush, S. Huiberts, and S. Tiwari, "On the integrality gap of binary integer programs with Gaussian data," *Mathematical Programming*, 2022, preliminary version in IPCO 2021. DOI: 10.1007/s10107-022-01828-1 arXiv: 2012.08346.
- [10] D. Dadush and S. Huiberts, "Smoothed analysis of the simplex method," in *Beyond Worst Case Analysis*, T. Roughgarden, Ed., Cambridge University Press, 2021, ch. 14, pp. 309–333. DOI: 10.1017/9781108637435.019 [Online]. Available: <https://sophie.huiberts.me/files/bwca-chapter-simplex.pdf>
- [11] D. Dadush and S. Huiberts, "A friendly smoothed analysis of the simplex method," *SIAM Journal on Computing*, vol. 49, no. 5, STOC18-449–499, 2020, preliminary version in STOC 2018. DOI: 10.1137/18M1197205 arXiv: 1711.05667.

Supervision

- Doctoral Researchers

- Eleon Bach. 2022-Present. TU Munich.
- Postdoctoral Researchers
 - François Lamothe. 2026-Present. LIMOS.
- Students
 - Guilherme Knupp Muniz. 2025. LIMOS.

--- Service to the community

- 2027 **Local Co-organizer**, *International Symposium on Mathematical Programming (ISMP)*
- 2026 **Organizer**, *Dagstuhl Seminar: Analysis of Algorithms Beyond the Worst Case*
- 2026 **Program Committee**, *Scandinavian Symposium on Algorithm Theory (SWAT)*
- 2026 **Program Committee**, *International Symposium on Theoretical Aspects of Computer Science (STACS)*
- 2025 – present **Editor**, *European Association for Theoretical Computer Science (EATCS) on Youtube*
- 2025 **PC Chair & Local Chair**, *Mixed Integer Programming Workshop Europe (MIP Europe)*
- 2025 **Program Committee**, *Caleidoscope Research School in Computational Complexity*
- 2025 **Program Committee**, *Mixed Integer Programming Workshop (MIP)*
- 2024 – present **Editorial Board**, *Discrete Optimization*
- 2024 **Program Committee**, *International Colloquium on Automata, Languages and Programming (ICALP)*
- 2024 **Program Committee**, *Integer Programming and Combinatorial Optimization (IPCO)*
- 2023 – present **Scientific Committee**, *Groupe de Travail Complexité et Algorithmes*
- Spring 2022 **Research Assessment Committee**, *Applied Mathematics TU Delft 2015–2020*
- May 2018 – Mar 2022 **Works Council**, *Centrum Wiskunde & Informatica*
Chair from March 2021 – March 2022.
- Nov 2017 – Jun 2022 **Local Representative**, *European Women in Mathematics The Netherlands*
- Aug 2020 – Dec 2020 **Seminar Organizer**, *Probability, Geometry, and Computation in High Dimensions Student Seminar*, Simons Institute for the Theory of Computing
- Oct 2019 – Apr 2020 **Communications Team**, *European Girls Mathematical Olympiad 2020*

--- Software

- Oct 2019 – Dec 2023 **Lead Author**, *LaTeX in Slack*
Mathematical remote collaboration software that served over 3500 simultaneous installs. Browser extension that allows $\text{\LaTeX}$ formulas on chat service Slack.

Miscellaneous writing

- *Het Dieetprobleem*, STAtOR, July 2025.
- *Prizes and Prejudice*, Bulletin of the European Association for Theoretical Computer Science, October 2022.
- *Wanneer is Software Snel?*, AG Connect, November 2021.

Presentations

Lecture Series

- *Open Problems about the Simplex Method*, IPCO Summer School, July 1–2, 2024.

Keynote Lectures

- *Open Problems about the Simplex Method*
 - CIAC, Rome, June 10–12, 2025.
 - Workshop on Polytopes, February 24–28th, 2025.
 - Combinatorics and Geometry in Villeteuse, November 7–8, 2024.
- *Smoothed Analysis of the Simplex Method: Upper and Lower Bounds*
 - Workshop Complexité et Algorithmes de CoA, September 19–20, 2023.
 - Mixed Integer Programming Workshop, May 22–25, 2023.
 - Aussois Combinatorial Optimization Workshop, January 9–13, 2023.

Contributed & Seminar Presentations

- *Beyond Smoothed Analysis: Analyzing the Simplex Method by the Book*
 - British Colloquia in Combinatorics, May 14th, 2026.
 - Discrete Optimization, Hausdorff Center for Mathematics, April 24th, 2026.
 - KTH Stockholm, March 24th, 2026.
 - MOSEK, March 17th, 2026.
 - VVSOR Annual Meeting, March 5th, 2026.
 - TCS+, March 4th, 2026.
 - Gurobi, February 18th, 2026.
 - ETH Zurich, November 18th, 2025.
- *Everything Wrong with Smoothed Analysis*, Aussois Combinatorial Optimization Workshop, January 5–9, 2025.
- *Linear Programming: Algorithms and Ethics*, ENS Rennes, December 3rd, 2025.
- *The Diet Problem*, G-SCOP, October 9th, 2025.
- *Optimal Smoothed Analysis of the Simplex Method*
 - FOCS, YouTube, December 2025.
 - OPTIMA, August 20th, 2025.
- *The Life and Work of Paula Harris*
 - Mixed Integer Programming, YouTube, August 2025.
 - The Simplex Method: Theory, Complexity, and Applications, June 26–27, 2025.
- *Open Problems about the Simplex Method*
 - UCLouvain, April 15th, 2025.
 - ETH Zurich, April 10th, 2025.
 - Séminaire du GT Graphes, March 20th, 2025.
 - LIRMM Montpellier, January 14th, 2025.
 - OMP, Antwerp, October 16th, 2024.
 - EPFL Bernoulli Center, August 19th, 2024.
 - The University of Edinburgh, May 22nd, 2024.
 - Séminaire Parisien d'Optimisation, May 6th, 2024.
 - Bocconi University, April 10th, 2024.
 - Journées SMAI-MODE, March 27–29, 2024.
 - Journées Nationales du GdR IFM, March 21st, 2024.
 - IRIF, February 27th, 2024.
 - Université Claude Bernard Lyon 1, February 12th, 2024.

- Zuse Institut Berlin, February 7th, 2024.
- FU Berlin, February 1st, 2024.
- 27th Seminar of the POC group, December 15th, 2023.
- CWI Networks & Optimization Seminar, November 15th, 2023.
- *A Nearly Optimal Randomized Algorithm for Explorable Heap Selection*
 - University of Washington, May 30th, 2023.
 - École Polytechnique, March 14th, 2023.
- *Smoothed Analysis of the Simplex Method: Upper and Lower Bounds*
 - SIAM Conference on Applied Algebraic Geometry, July 10–14, 2023.
 - Journées Polyèdres et Optimisation Combinatoire, June 28–30, 2023.
 - STOC, YouTube, June 2025.
 - Copenhagen-Jerusalem Combinatorics Seminar, May 4, 2023.
 - University of Massachusetts Amherst Theory Seminar, April 4th, 2023.
 - ICERM Combinatorics and Optimization Workshop, March 27–31, 2023.
 - CNRS LIMOS lab, March 9th, 2023.
 - Rutgers/DIMACS Theory of Computing Seminar, February 22nd, 2023.
 - CUNY Queens College CS Seminar, February 15th, 2023.
 - University of Michigan Theory Seminar, January 27th, 2023.
 - Groningen Probability and Statistics Seminar, December 15th, 2022.
 - EPFL Optimization Seminar, December 7th, 2022.
 - CNRS LIPN lab, December 1st, 2022.
 - NYU Theory Seminar, November 17th, 2022.
- *Combinatorial Diameter of Random Polytopes*
 - Discrete Optimization Talks, YouTube, October, 2022.
 - Columbia CS Theory seminar, October 7th, 2022.
 - Cargese Workshop on Combinatorial Optimization, September 23rd, 2022.
 - Workshop *Convex Geometry and its Applications*, MFO Oberwolfach, December 12–18 2021.
 - Symposium on Computational Geometry (SoCG), June 8th, 2022.
 - Online Asymptotic Geometric Analysis Seminar, September 28th, 2021.
 - TU Berlin DISCOGA Research Seminar, September 14th, 2021.
 - Dutch Optimization Seminar, May 27th, 2021.
 - Simons Institute Polymath Project Talks, December 10th, 2020.
- *Why Can We Solve (Integer) Linear Programs?*, STOC TCS Women Spotlight Workshop Rising Star Talk, June 22nd, 2021.
- *On the Integrality Gap of Binary Programs with Gaussian Data*
 - UC Davis Mathematics of Data & Decisions Seminar, February 23rd, 2021.
 - London School of Economics PhD Seminar on Combinatorics, Games and Optimization, February 5th, 2021.
- *Simple Iterative Methods for Linear Optimization over Convex Sets*
 - Integer Programming and Combinatorial Optimization (IPCO), June 27th, 2022.
 - CU Boulder CS Theory Seminar, February 2nd, 2021.
- *Layered-Least-Squares Interior Point Methods — a Combinatorial Optimization Perspective*
 - CWI N&O Online seminar, May 14th, 2020.
 - TU Munich Applied Geometry and Discrete Mathematics Seminar, March 3rd, 2020.
- *A Friendly Smoothed Analysis of the Simplex Method*
 - Utrecht University Algorithms Seminar, September 3rd, 2019.
 - *Computability in Europe* in Durham, July 15–19, 2019.
 - SIAM Conference on Applied Algebraic Geometry in Bern, July 9–13, 2019.
 - *Workshop on Polytopes* in Bochum, March 11–15, 2019.
 - *23rd Combinatorial Optimization Workshop* in Aussois, January 7–11, 2019.
 - Workshop *Convex Geometry and its Applications* at the Mathematisches Forschungsinstitut Oberwolfach, December 9–15, 2018.
 - *DIAMANT Symposium* in Veenendaal, November 29–30, 2018.
 - *Ninth Cargese Workshop on Combinatorial Optimization* in Cargese, October 15–19, 2018.
 - *International Symposium on Mathematical Programming* in Bordeaux, July 1–6, 2018.
 - *Symposium on Theory of Computation* in Los Angeles, June 25–29, 2018.

- *Highlights of Algorithms* in Amsterdam, June 4–6, 2018.